Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.

Final Answer: 23. Substitute $x = 5$: $f(5) = 4(5) + 3 = 23$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 5(2) + 4 = 14$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 6(2) + 5 = 17$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 24.5. The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(2)$.

Final Answer: 10. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for simplifying radical expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve $2^{x+1} = 5^{x-1}$, giving your answer to 3 significant figures.

Take logs: $(x+1)\ln 2 = (x-1)\ln 5$. $x\ln 2 - x\ln 5 = -\ln 5 - \ln 2$. $x = \frac{-\ln 10}{\ln 2 - \ln 5} = \frac{-\ln 10}{\ln(2/5)} \approx 2.48$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.

Step 1: Convert to exponential form: $x^2 - 7 = 3^2 = 9$. Step 2: Solve the quadratic equation: $x^2 = 9 + 7 = 16 \Rightarrow x = \pm 4$. Step 3: Verify the domain $x^2 - 7 > 0$: For both $x = 4$ and $x = -4$: $x^2 - 7 = 16 - 7 = 9 > 0$. Both roots are valid. The solutions are **$x = 4$** and **$x = -4$**.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · MediumAn ecologist models the population P of trout in a lake as $P(t) = 1400 - 35t$, where t is the number of years after 2015. What is the best interpretation of the number 35 in this model?

The model is in linear form $P(t) = mt + b$, where $b = 1400$ is the initial population in year 2015, and $m = -35$ is the rate of change. The negative sign indicates a decrease. Therefore, 35 represents the estimated decrease in the trout population for each one-year increase in time.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWhich ordered pair (x, y) is a solution to the system of inequalities $y > 2x - 3$ and $y \leq -x + 4$?

Test the candidate point $(1, 2)$:First inequality: $2 > 2(1) - 3 \Rightarrow 2 > -1$ (True).Second inequality: $2 \leq -(1) + 4 \Rightarrow 2 \leq 3$ (True).Since $(1, 2)$ satisfies both inequalities simultaneously, it lies within the feasible solution region.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

Every Digital SAT question carries equal raw weight within a module. The student should immediately flag Questions 19–20, solve Questions 21–22 in under 90 seconds using Desmos, and spend the remaining 7.5 minutes on Questions 19–20 without time pressure.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

The Digital SAT SPR allows up to 5 characters including negative signs and decimal points:As a fraction: -7/4 (4 characters).As a terminating decimal: -1.75 (5 characters).Invalid: Mixed number -1 3/4 is read as -13/4 and will be graded incorrect.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the x-intercept and y-intercept of the linear equation $3x - 6y = 18$.

x-intercept: Set $y = 0$: $3x - 6(0) = 18 \Rightarrow 3x = 18 \Rightarrow x = 6$. Intercept is $(6, 0)$. y-intercept: Set $x = 0$: $3(0) - 6y = 18 \Rightarrow -6y = 18 \Rightarrow y = -3$. Intercept is $(0, -3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Linear Equations & Lines

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

(a) Inner function first: $f(2) = 4(2) - 3 = 5$. Then $g(5) = 5^2 + 1 = 26$. (b) Inverse of g: Let $y = x^2 + 1 \Rightarrow x^2 = y - 1 \Rightarrow x = \sqrt{y - 1}$. Thus $g^{-1}(x) = \sqrt{x - 1}$ for $x \geq 1$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Rational Expressions

Level 3 · AdvancedFor the piecewise function $g(x) = \{2x + 5, x \leq -1; x^2 - 3, x > -1$, what is the value of $g(-1) + g(3)$?

Evaluate $g(-1)$: Since $-1 \leq -1$, use the first piece: $g(-1) = 2(-1) + 5 = 3$. Evaluate $g(3)$: Since $3 > -1$, use the second piece: $g(3) = 3^2 - 3 = 6$. Sum: $g(-1) + g(3) = 3 + 6 = 9$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Statistics & Data Analysis

Level 2 · MediumA study of 100 students recorded test results based on exam prep: • Studied: 54 passed, 6 failed (60 total) • Did not study: 18 passed, 22 failed (40 total) What is the probability that a randomly chosen student passed the exam, given that the student studied?

The condition 'given that the student studied' restricts the denominator to the 60 students who studied: $P(\text{Passed} \mid \text{Studied}) = \frac{54}{60} = 0.90 = 90\%$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

Level 4 · SynthesisIf $f(x-3)=4x+5$ for all real numbers x, what is the value of $f(2)$?

To evaluate $f(2)$, set the input expression $x-3$ equal to 2: $x-3=2 \Rightarrow x=5$. Substitute $x=5$ into the right-hand side: $f(2)=4(5)+5=25$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Level 2 · MediumLet $f(x)=x-4$ and $g(x)=2x^2+1$. If $h(x)=g(f(x))$, what is the value of $h(1)$?

Step 1: Evaluate inner function at 1: $f(1)=1-4=-3$. Step 2: Pass result into $g(x)$: $g(-3)=2(-3)^2+1=2(9)+1=18+1=19$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

Level 1 · FoundationsFor the function $f(x)=2x^2-3x+5$, what is the value of $f(-3)$?

Substitute $x=-3$ into the expression, taking care with negative signs:

$f(-3)=2(-3)^2-3(-3)+5=2(9)+9+5=18+9+5=32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x)=2x+8$ in multiplication properties of exponents, evaluate $f(2)$.

a. 12 ✓

b. 14

c. 4

d. 16

Substitute $x=2$: $f(2)=2(2)+8=12$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x)=3x+9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x=2$: $f(2)=3(2)+9=15$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplication properties of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5 **b.** 19.5

c. 20.5 ✓ **d.** 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5 **b.** 20.5

c. 21.5 ✓ **d.** 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. **20** ✓ **b.** 24

c. 32 **d.** 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$:
$f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19 ✓

b. 31

c. 16

d. 11

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$ ✓

b. $y = 0$

c. $y = 3$

d. $x = -1$

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the x-axis, forgetting the vertical shift by -5..

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. **a = 4, b = 3** ✓

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12 \cdot b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b must be positive for an exponen.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. **4** ✓

d. 6

Since the graph of $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$, we can set up a system of two equations: 1. $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2. $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponential function, $b > 0$. Thus, $b = 3$. Now, substitute the value of b back into equ.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. **8** ✓

c. 9

d. 5

The exponential function is given by $f(x) = 2^x$. To find the value of the function when $x = 3$, we substitute 3 for x into the function's formula: $f(3) = 2^3$ $2^3 = 2 \times 2 \times 2 = 8$ Therefore, the value of the function $f(x) = 2^x$ when $x = 3$ is 8.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. **8** ✓

c. 6

d. 16

To find the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$, we substitute $x = 3$ into the function: $f(3) = 2^{3+1}$ $f(3) = 2^4$ $f(3) = 16$ Therefore, when $x = 3$, $y = 16$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$ ✓

Applying Exponential & Logarithmic Functions principles, Standard half-life model. Therefore, the verified correct option is $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

a. $x = \ln k**2 + 1 - k$

b. $x = \ln \frac{k**2+1}{k}$

c. $x = \frac{\ln k**2+1}{k}$ ✓

d. $x = \frac{k}{\ln k**2+1}$

Applying Exponential & Logarithmic Functions principles, $2x = \ln 5 \Rightarrow x = \ln 5/2$. Therefore, the verified correct option is $x = \frac{\ln k**2+1}{k}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$ ✓

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

Applying Exponential & Logarithmic Functions principles, As $n \to \infty$: $A = Pe^{rt}$. Therefore, the verified correct option is $A = Pe^{rt}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

a. No simultaneous solution

b. $a = b$

c. $x = 0$

d. $x = a^b = b^a$ ✓

Applying Exponential & Logarithmic Functions principles, Both define x: $x = a^b$ and $x = b^a$. Therefore, the verified correct option is $x = a^b = b^a$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve: $5^x = 12$. Give exact answer.

a. $x = 12/5$

b. $x = \frac{\ln 12}{\ln 5}$ ✓

c. $x = \ln 12 - \ln 5$

d. $x = \log_{12} 5$

Applying Exponential & Logarithmic Functions principles, $x = \log_5 12 = \ln 12 / \ln 5$. Therefore, the verified correct option is $x = \frac{\ln 12}{\ln 5}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$ ✓

Applying Exponential & Logarithmic Functions principles, Change of base formula. Therefore, the verified correct option is $\frac{\ln x}{\ln b}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

The value of e to 4 decimal places is:

a. 2.1787

b. 3.1416

c. 2.6180

d. 2.7183 ✓

Applying Exponential & Logarithmic Functions principles, $e \approx 2.71828\ldots$ Therefore, the verified correct option is 2.7183.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Convert to exponential form: $\log_5 25 = 2$.

a. $5^{25} = 2$

b. $2^5 = 25$

c. $5^2 = 25$ ✓

d. $25^2 = 5$

Applying Exponential & Logarithmic Functions principles, Log to exponential. Therefore, the verified correct option is $5^2 = 25$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

Which is the y-intercept of $f(x) = 5 \cdot 3^x$?

a. $(1, 15)$

b. $(0, 15)$

c. $(0, 5)$ ✓

d. $(0, 3)$

Applying Exponential & Logarithmic Functions principles, $f(0) = 5 \cdot 1 = 5$. Therefore, the verified correct option is $(0, 5)$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4 ✓

d. 10

Applying Exponential & Logarithmic Functions principles, $2^{10/5} = 2^2 = 4$. Therefore, the verified correct option is 4.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Which of these is equivalent to $e^{2\ln 3}$?

a. $2e^3$

b. 9 ✓

c. 6

d. $\ln 9$

Applying Exponential & Logarithmic Functions principles, $e^{\ln 9} = 9$. Therefore, the verified correct option is 9.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Factoring Polynomials

Find the derivative of $f(x) = x^x$.

a. $x^x \ln x$

b. x^{x+1}

c. $x \cdot x^{x-1}$

d. $x^x(1 + \ln x)$ ✓

Logarithmic differentiation: $\ln y = x \ln x \Rightarrow y'/y = 1 + \ln x \Rightarrow y' = x^x(1 + \ln x)$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

Find the number of solutions to $e^x = x + 2$.

a. 1

b. 2 ✓

c. Infinitely many

d. 0

Graph: e^x convex, $x + 2$ linear — they intersect at 2 points for this particular line.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

$x \ln 3 = (x+1)\ln 2 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

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